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What Does Everybody Know?

How a Simple Question Revealed the Common Language of Fire Alarm Systems

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A fire alarm system reports a condition. Someone responsible for the building wants to understand what happened, where it happened, and what to do next. They do not need to begin with the manufacturer's communication protocol. They need meaning.

WHAT HAPPENED?	WHERE?	WHAT SHOULD I DO?

A question behind the software

When I began developing software for fire alarm systems in the 1980s, my motivation was practical. I wanted one application to work with as many manufacturers as possible without rewriting everything for each new connection.

Looking back, I think I may have been asking myself a deceptively simple question: "What does everybody know?"

Not that everybody knows the same facts. A guard, firefighter, technician, engineer, and building owner bring different experience. But nearly everyone recognizes the questions that need answering when something important happens.

The panel manufacturer had its own way of reporting. The person receiving the report had ordinary questions. What mattered was finding the information that could connect the two.

> **The questions were familiar. The answers arrived in different dialects.**

Different reports. Familiar information.

Look at these selected excerpts from fire alarm messages in the RescueLogic manufacturer comparison. They are different example events, not competing reports about one device. Their layouts, abbreviations, and vocabulary differ, but recognizable information keeps appearing.

Source example	Sample reported message excerpt

Kidde-Fenwal	`12:00:00  04/14/25`
	`N01-L1:001 ALARM ON PHOTO`
	`CONFERENCE ROOM`

Fike	`ALARM: PHOTO DETECT`
	`CONFERENCE ROOM  1-001`
	`17:27:46 04/14/2025`

Potter (XML)	`<state>Alarm</state>`
	`<ptname>CONFERENCE ROOM</ptname>`
	`<ptid>009</ptid>`
	`<pttype>SMOKE DETECTOR</pttype>`

Five roles underneath the differences

What people want to know	The information role
Who is reporting?	Reporting system / panel
Which point?	Device or point identity
What kind of point?	Device type
Where does it belong?	Device location
What is it reporting?	Reported condition

These are information roles, not five fields guaranteed to appear in every transmitted message. Since the meaning exists at the point level, the customer database may sometimes supply context absent from the source report.

A simple first level of recognition

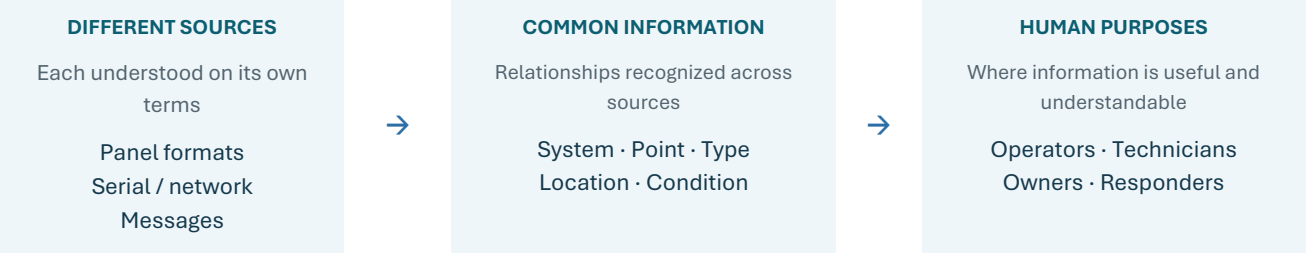
In a simplified human-facing view, three familiar condition categories can help organize attention. The device type, the precise report, and the site's approved procedures still determine the appropriate response.

NORMAL	ACTIVE	TROUBLE
No abnormal condition reported. Continue appropriate monitoring.	A condition needing attention is reported. Follow the applicable plan.	A fault or impaired system function is reported. Investigate under the applicable procedures.

These three teaching categories are not a definition of SAFER's six numeric STATUS reporting codes, but they correlate. A received report can describe a restoration, repetition, or status observation rather than a new physical transition.

The connection became an architecture

Looking back, I learned to understand each manufacturer's reporting system on its own terms. Its messages, terminology, and reporting behavior were unique, but as I understood the differences, familiar relationships became visible. Cadgraphics developed manufacturer-specific interfaces to interpret the reports and relate them to a common set of installation records. The common structure emerged from understanding the differences, rather than expecting the equipment to be identical.



A simple view of the architecture: the sources can vary, while the roles used to understand a report remain recognizable.

Common expectations make differences recognizable

Standards help establish common expectations. NFPA 72 provides requirements for the installation, performance, inspection, testing, and maintenance of fire alarm systems. Those expectations give us a basis for evaluating an installation. A common information model provides another useful basis for comparison: it allows people to recognize when reported device information, maintained records, and drawings do not agree. Recognizing a difference is the beginning of verification, not proof that something is wrong.

A practical result, not a claim of identical equipment

Over decades, RescueLogic connected with different manufacturers using dedicated interfaces and, later, configurable interpretation rules. The customer records did not need to be reinvented every time the connection method or product name changed.

The Script Panel made many interpretations adjustable by trained maintainers. The July 2016 Marriott demonstration brought several manufacturers into a shared RescueLogic environment. These are examples of the approach in practice, not evidence that every possible connection requires no special engineering.

Turning the architecture outward

Originally, the challenge was to let one application understand reports from many manufacturers. The same underlying information can now support more than one application: a responder’s view, a technician’s test record, an owner’s maintained drawings, or an interface to another software system.

SAFER is a later proposal for carrying reported event meaning in a structured JSON payload. It standardizes the proposed message content, but does not require changes to the fire alarm equipment, communication pathway, cybersecurity, or code-compliance responsibilities.

> **Many manufacturers came into one model.**
> **Now one model can open outward into many human purposes.**

An unanswered question still has a place

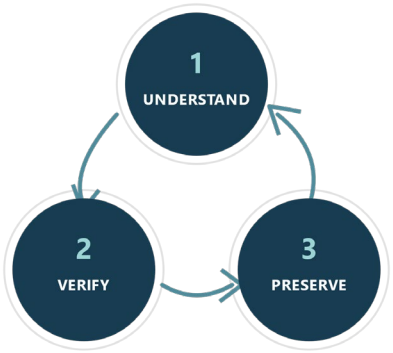
Several devices may share a description such as First Floor without sharing the same identity. Within RescueLogic, each point has a defined relationship to its reporting panel, device type, and assigned location. These relationships help prevent conflicting assignments and make discrepancies easier to recognize.

For example, if two physical devices use the same address on the same panel loop, the panel may report a Duplicate Address condition. That message gives a technician something specific to investigate. By itself, it is a clue, but does not establish which device or record needs correction.

Blank fields and conflicting values are also opportunities for investigation. We can preserve the information already established without inventing missing details. Once the correct information has been verified, it can be added to the maintained record.

> **An unknown is not an error to conceal. It is an open question with a legitimate place in the information model.**

Three interlocking principles



01 UNDERSTAND
Recognize what belongs together.

02 VERIFY
Investigate what does not agree.

03 PRESERVE
Keep what has been established.

Each reinforces the others.

Understanding makes information recognizable. That understanding helps people notice differences and investigate them. Verification improves the information, and preservation makes the results available to the next person. Each supports the others. Preserved knowledge becomes the starting point for further understanding, comparison, and correction.

The questions were always there

What happened? Where did it happen? What should I do? Those are human questions. The equipment contributes reports; a maintained information model connects them to devices and places; people verify what is true and decide what to do.

We started with the questions people had in common, discovered the information the systems had in common, and built a practical connection between the two.

When knowledge is understandable, people can help verify it. When verified knowledge is preserved, the next person can build upon it rather than begin again.

COMPANION READING One Architecture. Many Manufacturers. Decades of Continuity. (October 2026); Fire alarm panels speak the same language; The Grammar of a Fire Alarm System; The Ordered Fire Alarm System; SAFER Protocol v1.1. SAFER review materials: github.com/rescuelogic/SAFER